

# Hardware Flow

Smart Cold-Chain System — Farm-to-Market Closed Cold-Chain | Physical Architecture & Component Flow

## 1. Overview

The hardware architecture spans three physical zones: the Central Cold-Storage Facility (grid-powered refrigeration + solar-assisted crate charging), the Intelligent Transport Crate (a self-contained sensing, control and actuation unit), and the Vehicle/Field layer (mobile connectivity and location tracking). The same physical crate moves across all three zones without the produce ever being transferred to a different container.

## 2. Hardware Architecture Layers

| Layer           | Function                            | Key Components                                                              |
|-----------------|-------------------------------------|-----------------------------------------------------------------------------|
| Sensing         | Measure environmental conditions    | SHT31 (temperature + humidity), warehouse ambient sensors                   |
| Control         | Process sensor data, decide actions | ESP32-C3 microcontroller                                                    |
| Actuation       | Maintain required temperature       | TEC / Peltier thermal-maintenance unit                                      |
| Power           | Supply and store energy             | LiFePO4 battery + BMS, solar panels, grid refrigeration, PCM thermal buffer |
| Communication   | Send live data to backend           | Wireless/GSM/IoT communication module, GPS                                  |
| Display & Alert | Local status indication             | LCD/OLED screen, LED indicator, buzzer                                      |

## 3. Central Cold-Storage Facility — Hardware

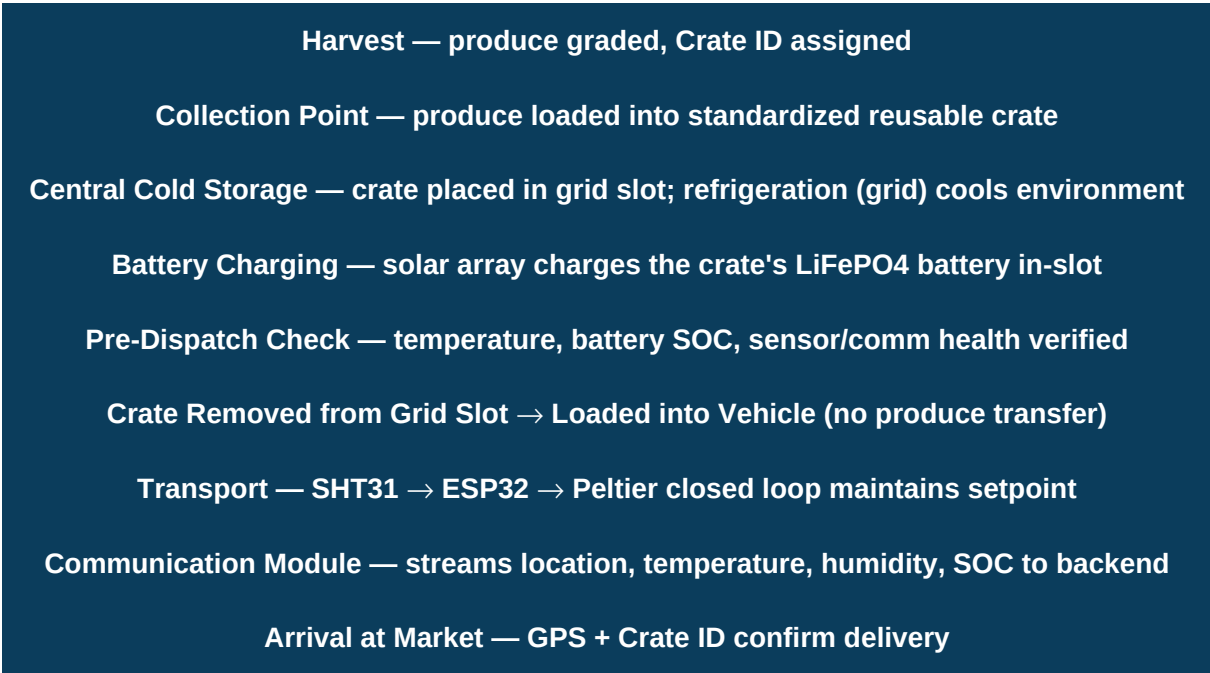
- **Grid-based storage racks:** Physical slots/positions, each mapped to a unique grid ID for crate placement.
- **Central refrigeration unit:** Grid-powered compressor/cooling system maintaining warehouse-wide temperature and humidity.
- **Solar panel array + charge controller:** Dedicated to charging individual crate batteries while crates are stored (not warehouse refrigeration, under normal operation).
- **Hybrid inverter:** Bridges solar and grid supply; used to route solar power to refrigeration only during a grid-outage emergency mode.
- **Warehouse environmental sensors:** Fixed sensors monitoring storage-block temperature, humidity and power status.
- **PCM (Phase-Change Material) thermal buffers:** Passive thermal storage in cold-storage blocks to slow temperature rise during a total power outage.
- **Facility gateway/server:** Local hub aggregating crate and warehouse sensor data before uplink to the cloud backend.

## 4. Intelligent Transport Crate — Bill of Materials

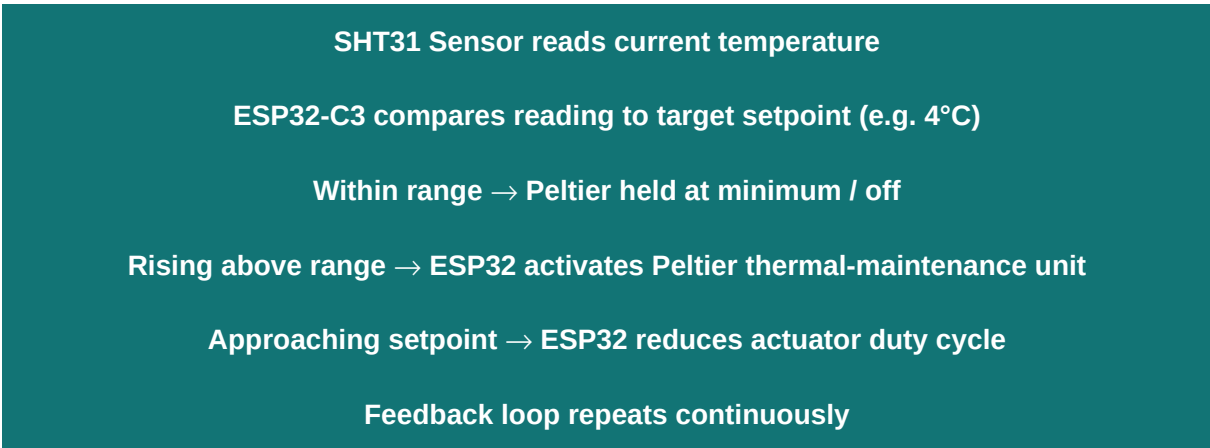
| Component          | Part                               | Role                                                   |
|--------------------|------------------------------------|--------------------------------------------------------|
| Thermal Insulation | PU / PIR rigid foam                | Minimizes heat ingress; reduces actuator load          |
| Sensor Unit        | SHT31                              | Reads internal temperature and humidity                |
| Controller         | ESP32-C3                           | Runs closed-loop control logic and communication       |
| Actuator           | TEC / Peltier module               | Corrects temperature deviation on demand               |
| Power              | LiFePO4 battery + BMS              | Powers all crate electronics; BMS protects/reports SOC |
| Local Display      | LCD / OLED + LED / buzzer          | On-crate status and abnormal-condition indication      |
| Tracking           | Unique Crate ID tag + comms module | Identity, GPS location, and data uplink                |

Design note: the storage crate and the transport crate are the **same physical unit** — it is lifted out of its warehouse grid slot and placed directly into the vehicle, eliminating a produce-transfer step and its associated cold-chain break risk.

## 5. End-to-End Hardware Flow (Farm to Market)



## 6. In-Transit Closed-Loop Control (Hardware Signal Path)



The PU/PIR insulation passively limits heat gain so the Peltier only needs to supply the correction, not continuous full-power cooling — this is the main hardware-level battery-saving design.

## 7. Power-Source Hardware Behaviour

| Scenario                             | Refrigeration Hardware Path                                           | Crate Battery Charging Path                                |
|--------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------|
| Grid ON, Solar OFF (normal / cloudy) | Grid → central refrigeration                                          | Grid → dedicated battery charger                           |
| Grid OFF, Solar ON                   | Solar → hybrid inverter → refrigeration (emergency)                   | Solar → crate battery (continues)                          |
| Grid OFF, Solar OFF                  | No active refrigeration — PCM + insulation hold temperature passively | No charging — dispatch crates with sufficient existing SOC |

## 8. Pre-Dispatch Hardware Checklist

- Produce temperature already at required setpoint (crate does not rapidly cool warm produce).
- Battery SOC sufficient for the estimated journey duration.
- Estimated temperature-maintenance runtime > expected journey time.
- SHT31 sensor, ESP32 controller and communication module all responding.
- Crate physically sealed and insulation intact.
- For hilly/remote routes: lighter vehicle selected, fewer crates per trip, battery runtime re-verified against route travel time.